From Impact Scan to Actionable Interventions

A Transparent Decision Support Tool for Heijmans Vastgoed

Team 11 – Intervention Recommender Gilbert Laanen Rick de Rijk
Stefan Vonk Tycho van Rooij Thom Verzantvoort Patryk Grodek

2026-05-26

Table of contents

1 Executive Summary **3**

2 Client Context and Problem Understanding **3**

3 Analytical Strategy and Methodological Justification **4**
3.1 Decision-Support Approach and MCDM Rationale 4
3.2 Scoring Framework 4
3.3 Theme Remapping: Integrating Heijmans Interventions into the Existing Database 5
3.3.1 Caveats and Limitations 6
3.4 Selection and Deduplication of Heijmans Interventions 6
3.5 Intervention-Category-Theme Mapping for New Interventions 7

4 Data Understanding, Preparation, and Quality **7**

5 Analysis Execution and Results **8**
5.1 Scoring Method 8
5.2 Recommender Output & Interpretation 9
5.3 Case-Based Interpretation 9
5.4 From Ranking to Decision 10

6 Why the Scores Are Trustworthy **10**

7 Recommendations **10**

8 Limitations **11**

9 Conclusion **11**

Appendix A: Overlap Analysis — Excluded Interventions **12**

Appendix B: Scoring Rationale - Initial Batch **13**

Appendix C: Scoring Rationale - Heijmans Interventions **14**

Appendix D: Interventiecategorie to Database Category Mapping **24**

References **25**

List of Figures

1 Recommender output showing ranked interventions 9

1 Executive Summary

Heijmans Vastgoed aims to systematically improve livability and well-being in residential developments. To this end, the organization already possesses a strong analytical instrument in the form of a well-being impact scan, which identifies neighborhood challenges across social, spatial, and environmental dimensions.

However, the consultancy process currently breaks down after initial analysis. While the scan diagnoses problems effectively, translating insights into concrete interventions is currently inconsistent across comparable projects, highly time-consuming, difficult to explain to stakeholders, and highly dependent on individual consultants.

The core issue is not a lack of data, but the absence of a structured mechanism that converts analytical insight into actionable decisions. As a result, similar neighborhood profiles can lead to different recommendations, with limited transparency on how those decisions are made.

This project addresses that gap by introducing a decision-support recommender system that systematically prioritizes interventions based on contextual relevance and implementation feasibility, bridging the gap between analysis and intervention selection. The system combines a structured intervention database with a transparent scoring model that makes trade-offs between impact, cost, risk, and effort explicit.

The recommender does not replace Heijmans' consultants. Rather, it enables consultants to move from analysis to decision-making in a consistent and explainable way. Rather than prescribing a single solution, it structures the decision space and makes underlying assumptions transparent and reproducible.

In terms of business value for Heijmans, this results a faster advisory workflow, more consistent intervention recommendations across consultants and projects, improved explainability towards municipalities and other stakeholders, and a scalable approach to intervention selection.

2 Client Context and Problem Understanding

Heijmans Vastgoed operates in complex urban development environments involving municipalities, housing corporations, investors, and residents. Their projects must generally find a balance between long-term livability goals, financial feasibility, regulatory constraints, stakeholder alignment, implementation risk, and other aspects.

The current well-being impact scan provides valuable insights and a structured overview of neighborhood performance across multiple themes.

However, while the scan identifies where challenges exist, it does not specify which interventions should be implemented. In practice, consultants manually translate these insights into recommendations.

This creates three main operational shortcomings: - Inconsistent outcomes: neighborhoods with similar profiles can produce different recommendations - Limited explainability: it is difficult for stakeholders to trace back why a particular intervention was chosen over other options. - Inefficient knowledge re-use: the manual nature of the impact scan keeps valuable knowledge to individual consultants, rather than embedded organizationally where the knowledge could be re-used to speed up the decision-making process.

The underlying problem is therefore not analytical, but translational. The organization lacks a structured mechanism that connects insight to action in a consistent and explainable way.

Therefore, our objective is to design a system that can translate impact scans into a consistent set of recommended interventions, based on explainable and standardized factors. The system should support consultants in their decision-making process, not fully replace them, and should be scalable across projects.

3 Analytical Strategy and Methodological Justification

The objective of this project is not to improve analytical diagnosis but to improve decision-making following the well-being impact scan. The analytical strategy therefore focuses on structuring how insights are translated into interventions rather than developing predictive models or discovering new patterns in the data.

In consultancy practice, intervention selection is inherently a multi-criteria decision problem. Consultants must simultaneously consider thematic relevance, expected impact, financial implications, implementation effort, feasibility within local governance structures, and risk exposure. These considerations are typically evaluated implicitly through professional judgment. While effective at the individual level, implicit reasoning limits consistency, transparency, and organizational learning.

To address this, the solution was designed as a decision-support recommender grounded in Multi-Criteria Decision Making (MCDM). The guiding principle behind the methodological design was interpretability. Recommendations must remain understandable to consultants, clients, and external stakeholders, meaning methodological transparency was prioritized over algorithmic sophistication.

3.1 Decision-Support Approach and MCDM Rationale

The recommender follows a two-phase structure. First, a constraint-filtering stage removes interventions that are incompatible with requirements defined for the project context. After all, options that cannot realistically be implemented should not enter the evaluation process. By separating feasibility constraints from preference evaluation, the model prevents unrealistic interventions from influencing rankings.

After this filtering stage, the remaining interventions are evaluated using a Weighted Sum Model (WSM), also referred to as Simple Additive Weighting. Each intervention is assessed across multiple criteria which are aggregated into a single interpretable score through weighted linear combination. The approach allows every component of the final recommendation to be traced back to explicit assumptions about priorities and trade-offs.

This separation between hard constraints and soft preferences is well-established in decision-support literature and ensures that only feasible options are considered before trade-offs are evaluated (Domshlak et al. 2006; Abdelli et al. 2020; Khadra et al. 2020).

The selection of WSM over alternative methods such as AHP or TOPSIS is motivated by three factors. Firstly, WSM provides the highest level of transparency of these methods, as its scores are directly traceable to individual criteria. Given the importance of this traceability in a consultancy context, it is a benefit that cannot be ignored. Secondly, WSM scales more efficiently for large sets of interventions. As scalability is one of the objectives of the recommender system, this too is a significant argument in WSM's favor. Lastly, literature shows that the difference in results is often minimal between simple linear models such as WSM and more complex implementations (Ciardiello and Genovese 2023; Marinoni 2006; Vagiona 2021).

The choice of WSM therefore represents a deliberate trade-off. The model sacrifices some mathematical sophistication in favor of usability, scalability, and explainability. These characteristics align closely with Heijmans' operational context, where recommendations must be defendable in collaborative decision environments involving municipalities, project partners, and residents.

The analytical strategy thus positions the recommender not as an optimization engine but as a structured reasoning framework that externalizes consultant judgment and makes decision logic explicit.

3.2 Scoring Framework

The scores assigned to each intervention reflect a manual, domain-informed review by the consulting team, replacing the synthetic AI-generated baseline. For each intervention, seven dimensions are scored on a scale of 1 to 5. For impact potential and feasibility, higher scores are better. For implementation effort,

implementation risk, and cost, higher scores indicate worse outcomes, and these dimensions are inverted internally by the recommender. Measurability and data availability are informational dimensions that inform confidence in the recommendation but do not directly penalise or reward an intervention in the ranking.

The seven dimensions are defined as follows. Impact potential reflects the degree to which the intervention is expected to move the needle on wellbeing outcomes at neighbourhood level, where a score of 1 indicates marginal or highly indirect impact and a score of 5 indicates strong, area-wide, durable impact across multiple wellbeing dimensions. Implementation effort captures the operational complexity of delivering the intervention, including coordination requirements, planning processes, and personnel demands, where higher means harder. Implementation risk reflects the likelihood that the intervention fails to deliver its intended outcome due to political, social, technical or institutional factors, where higher means riskier. Cost captures the financial investment required to implement the intervention at neighbourhood scale, where higher means more expensive. Measurability reflects the degree to which the outcome of the intervention can be observed and tracked using available monitoring tools, where higher means more measurable. Data availability reflects the degree to which baseline and outcome data exists at national or regional level to support the intervention, where higher means better data availability. Feasibility is a composite judgement of how realistic it is to implement this intervention across the range of municipalities and contexts in which Heijmans operates, where higher means more feasible.

For the initial batch of 44 interventions, scoring rationale has been documented for a representative selection, available in Appendix B. For all 46 interventions added from the Heijmans list, full scoring rationale is documented in Appendix C.

3.3 Theme Remapping: Integrating Heijmans Interventions into the Existing Database

The intervention database was constructed prior to the receipt of the Heijmans intervention list. Given the project timeline and the degree to which the database architecture was already developed, including scoring frameworks, ranking logic, and intervention metadata, a full reconstruction of the taxonomic structure was not feasible. The decision was therefore made to retain the existing 11 database themes as the authoritative classification framework and to map all incoming Heijmans interventions onto this structure.

The Heijmans intervention list organises interventions according to its own internal theme taxonomy, referred to as Impactthema's. These do not correspond one-to-one with the database themes, as they were developed independently and reflect Heijmans' internal project classification logic rather than the wellbeing dimension framework underpinning this database. To integrate the two, a systematic mapping was constructed in which each Heijmans Impactthema was assigned to the most conceptually proximate database theme. The mapping is presented in Table 1.

Table 1: Heijmans Impactthema to Database Theme Mapping

Heijmans Impactthema	Database Theme	Rationale
Sociale samenhang	THM_08 Sociale Samenhang	Direct conceptual equivalent
Gezond gedrag en gebruik	THM_07 Gezonde Leefstijl	Both concern health behaviour, physical activity, and lifestyle
Fysieke veiligheid	THM_02 Leefbaarheid	Physical safety is a constituent dimension of liveability in the Dutch urban planning context
Sociale veiligheid	THM_02 Leefbaarheid	Social safety is treated as a liveability dimension within the database construct
Klimaat & Milieu	THM_03 Milieukwaliteit & Klimaatadaptatie	Direct conceptual equivalent

Heijmans Impactthema	Database Theme	Rationale
Groen & Natuur	THM_04 Aantrekkelijke Groene Omgeving	Direct conceptual equivalent
Voorzieningen & Bereikbaarheid	THM_05 Toegang tot Voorzieningen & Mobiliteit	Direct conceptual equivalent
Uitnodigend en toegankelijk voor iedereen	THM_10 Diversiteit & Inclusie	Both concern accessibility, social inclusion, and removing barriers to participation
Perspectief	THM_11 Opgroeien & Ontwikkelen	Covers talent development and educational opportunity, aligning with the developmental focus of THM_11

This mapping was applied consistently across all interventions using the Impactthema field as the sole basis for theme assignment. The more granular Interventiecategorie field was deliberately excluded as a mapping criterion. While this field occasionally suggests an alternative theme assignment, applying it selectively would introduce inconsistency into the mapping logic and undermine the systematic nature of the approach.

A small number of interventions in the Heijmans list carry no Impactthema value. For these, theme assignment was made on the basis of the intervention content itself, and this is documented explicitly per intervention.

3.3.1 Caveats and Limitations

Three database themes have no direct Heijmans Impactthema equivalent: THM_01 Woonkwaliteit, THM_06 Trots & Identiteit, and THM_09 Zeggenschap & Participatie. As a consequence, this batch of Heijmans interventions does not materially reduce the underrepresentation of these themes in the database. THM_01 receives a small number of additions from the blank-Impactthema interventions where content clearly points to housing quality. THM_06 and THM_09 remain underrepresented and this constitutes a known limitation of the current database version. Addressing this would require either a targeted search for interventions explicitly covering identity, pride, and civic participation, or a future reassessment of the taxonomic structure if the project timeline permits.

3.4 Selection and Deduplication of Heijmans Interventions

The Heijmans intervention list contains 121 entries across three priority tiers, scored on a scale of 1 to 3 where 3 represents the highest organisational priority. The decision was made to include only interventions rated at priority level 2 or 3, yielding an initial candidate pool of 54 interventions. Priority level 1 interventions were excluded on the grounds that they represent lower organisational relevance from Heijmans' own assessment, and including them would introduce a large volume of poorly evidenced entries that would inflate the database without meaningfully improving its analytical coverage or quality.

Following this initial filter, a systematic overlap analysis was conducted to identify interventions in the candidate pool that were already substantively covered by existing database entries. The rationale for this step is that adding near-duplicate interventions creates redundancy in the ranking model and dilutes the discriminatory power of the recommendation engine. Two categories of overlap were identified. The first category comprises interventions where the Heijmans entry and an existing database intervention are conceptually equivalent, meaning they describe the same mechanism, target the same outcome, and operate at the same level of abstraction. These were excluded outright. The second category comprises interventions where the Heijmans entry constitutes a specific sub-component or single implementation measure of a broader existing database intervention. These were also excluded, on the grounds that the

existing intervention already captures this territory at the appropriate level of generality for the database construct.

Following the overlap analysis, 10 interventions were excluded from the candidate pool, resulting in a final addition of 46 new interventions. The total database size increases from 44 to 90 interventions. The full exclusion list with reasoning is documented in Appendix A.

3.5 Intervention-Category-Theme Mapping for New Interventions

To integrate the 46 new interventions into the recommender engine, each intervention required a corresponding entry in the intervention-category-theme mapping table, which serves as the operational link between an intervention and the thematic and categorical dimensions used for scoring and ranking. Two fields required explicit assignment decisions: the database category identifier and the theme match rating.

Category assignment was derived from the Interventiecategorie field in the Heijmans source file, which classifies each intervention at a more granular level than the Impactthema. Each unique Interventiecategorie value was mapped to the most conceptually proximate database category from the nine available categories, as documented in Appendix D. For example, interventions classified under Beweegroutes were assigned to CAT_04 Mobiliteit en toegankelijkheid, interventions under Kunst & Cultuur were assigned to CAT_08 Identiteit en cultuur, and interventions under Zeggenschap bewoners were assigned to CAT_07 Governance en participatie. This mapping was applied consistently across all 46 interventions.

All 46 interventions were assigned a mapping role of Primary, reflecting that the theme assignment represents the principal thematic fit for each intervention as defined by Heijmans themselves.

Theme match ratings were assigned based on the nature of the underlying theme mapping. For interventions whose Heijmans Impactthema constitutes a direct conceptual equivalent of the corresponding database theme — covering THM_07, THM_08, THM_03, THM_04, THM_05, THM_10, and THM_11 — a theme rating of 5 out of 5 was assigned, reflecting high confidence in the thematic fit. For interventions mapped to THM_02 Leefbaarheid via the Fysieke veiligheid or Sociale veiligheid Impactthema's, a theme rating of 4 out of 5 was assigned. These mappings are conceptually sound but represent a translation across non-equivalent taxonomies rather than a direct match, and the reduced rating reflects this additional degree of interpretive distance. Category ratings and combination match ratings were set at 4 for all 46 interventions, consistent with the fact that category assignments were derived from the Interventiecategorie field rather than from a purpose-built scoring exercise against the database category framework.

4 Data Understanding, Preparation, and Quality

The recommender relies on a structured intervention database that formalizes organizational knowledge about possible well-being interventions. In this database, each intervention is assigned relevant themes and categories based on provided impact scan data, as well as a set of evaluative characteristics describing the intervention's expected contributions and implementation burdens.

Interventions are scored on a five-point ordinal scale across dimensions reflecting impact, feasibility, cost, effort, and risk. Impact and feasibility represent benefit-oriented criteria, while cost, effort, and risk capture operational constraints associated with implementation. Currently, these scores are primarily interpolated using any available relevant data available alongside structured personal intuition on our part. It is up to Heijmans to apply their expert knowledge to obtain more accurate values for these scores.

Before analysis, scores are transformed to ensure comparability across criteria. All values are normalized to a common scale between zero and one, allowing criteria measured on different conceptual dimensions to be aggregated meaningfully. Burden-oriented criteria are inverted so that higher values consistently represent more desirable outcomes. This transformation simplifies interpretation and ensures that resulting rankings

remain intuitive.

Unlike predictive analytics projects, the dataset does not contain empirical outcome measurements. Instead, it represents structured expert knowledge accumulated through professional experience. This distinction is important: the recommender supports comparative decision-making rather than forecasting real-world impact.

The use of expert judgment introduces unavoidable subjectivity. Feasibility assessments may vary across municipalities, cost expectations may shift over time, and perceived risks may depend on local political conditions. Rather than attempting to eliminate subjectivity, the model makes these assumptions visible. Transparency becomes a mechanism for quality control, allowing consultants and stakeholders to discuss and refine evaluations collaboratively.

Data quality therefore depends less on statistical properties and more on governance practices. Maintaining the intervention database requires periodic review, calibration between consultants, and documentation of scoring rationales. Over time, organisational learning can be incorporated by updating scores based on project experiences, gradually strengthening the empirical foundation of the system.

Ethically, the approach avoids automated decision-making in contexts that affect lived environments. The recommender structures professional judgment but does not replace it, preserving human accountability in urban development decisions.

With the data structure defined, the next step is to examine how these elements are combined in the recommender to produce ranked outputs.

5 Analysis Execution and Results

5.1 Scoring Method

The analysis combines contextual relevance and implementation feasibility into a unified evaluation framework. Each intervention receives two intermediate scores. The fit score reflects how well an intervention aligns with the themes identified in the well-being impact scan, while the operational score captures the practical difficulty of implementation.

These dimensions are combined into a final score using weighted aggregation. In the default configuration, contextual fit receives greater emphasis than operational considerations, as addressing the challenges found in a neighborhood is considered the primary objective, with feasibility constraints still carrying secondary importance.

Recognising that projects differ in priorities, the system also provides other predefined weighting profiles. These include a budget-constrained option with high weight attributed to the cost score, a configuration with stronger emphasis on the relevant category over the theme(s), one that maximizes impact score, and lastly a "quick win" configuration that prioritizes feasibility and effort scores.

These profiles allow consultants to adapt the evaluation to situations such as strict budget limitations, urgency of implementation, or strategic emphasis on long-term impact. Should it be necessary, a fully configurable option where all weights can be changed manually is also available for advanced cases where project conditions require tailored decision logic.

The normalization approach, as described previously, ensures stability of results. Because scores are independent of the size of the intervention database, the addition of new interventions does not distort existing rankings, preserving interpretability over time. (Data Consultancy Team 2026; Marinoni 2006).

5.2 Recommender Output & Interpretation

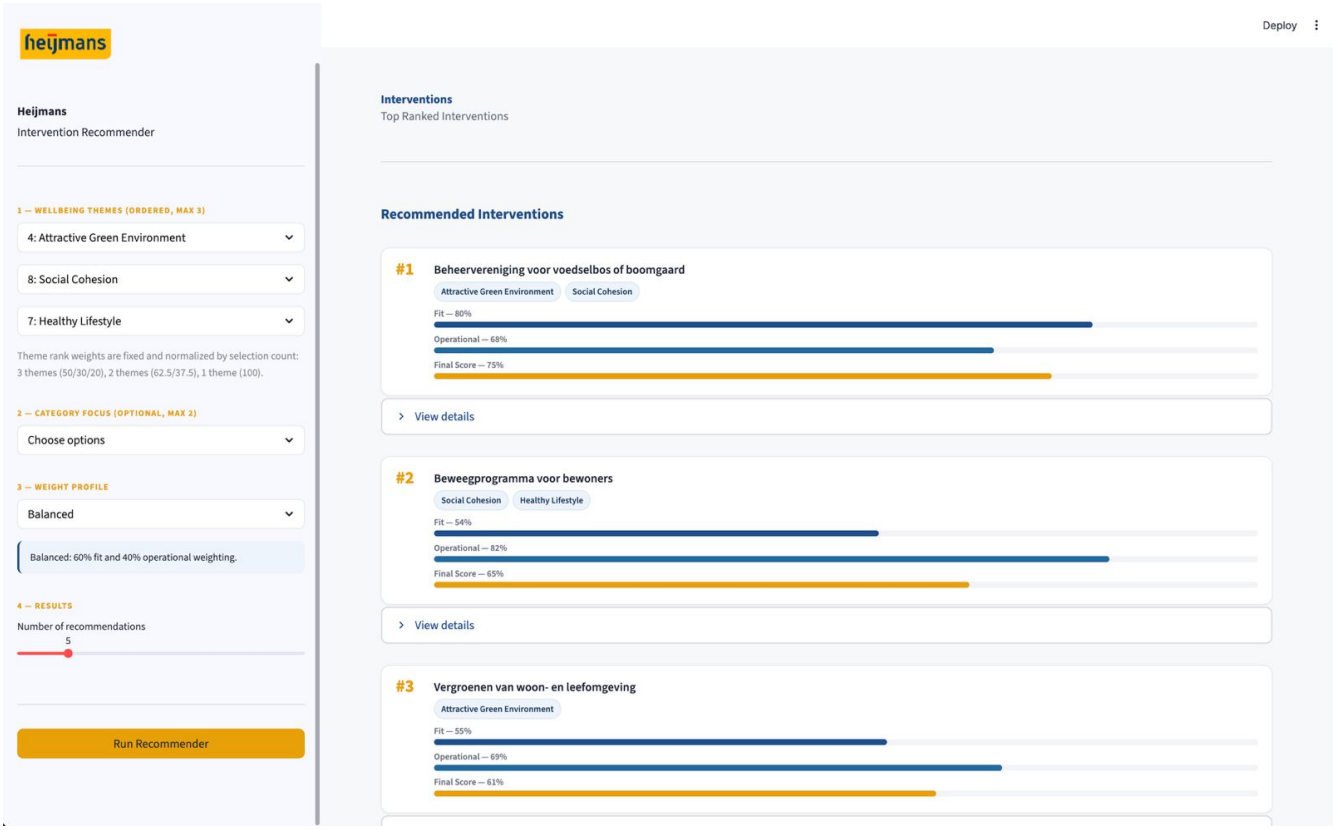


Figure 1: Recommender output showing ranked interventions

The recommender produces a ranked list rather than a deterministic answer. The ranking reveals how different interventions balance competing objectives. Three scores are presented per intervention: fit score, operational score, and final score.

The fit score reflects alignment with selected themes: how well the intervention is projected to resolve the issues found in the neighborhood. The operational score reflects implementation feasibility, showing to what degree the intervention aligns with the constraints. The final score then combines both perspectives for a final single rating for each intervention.

High-impact interventions may rank lower when associated with substantial implementation effort or risk, while relatively modest interventions can rise in the ranking due to strong feasibility.

This behavior is expected for a realistic decision-making process. Urban interventions rarely optimize a single dimension; successful recommendations emerge from negotiated trade-offs between ambition and practicality.

The output therefore shifts discussion from selecting interventions based on intuition toward evaluating explicitly defined trade-offs. Consultants can explain not only which intervention is recommended, but why alternatives were ranked differently.

5.3 Case-Based Interpretation

In practical use, the highest-ranked intervention represents the strongest overall balance between relevance and feasibility given the chosen priorities. Lower-ranked interventions should not be interpreted as poor options. Instead, they represent alternatives that become attractive under different constraints, such as limited budgets or accelerated timelines.

The recommender thus expands rather than narrows decision space. It enables consultants and clients to explore scenarios and understand how changing priorities affects outcomes, strengthening collaborative decision-making.

5.4 From Ranking to Decision

Importantly, the recommender does not make decisions. Its purpose is to structure reasoning. Consultants remain responsible for interpreting results, validating contextual assumptions, and assembling final advisory recommendations. The system reduces cognitive complexity while preserving professional discretion.

It is important to keep in mind that the recommender provides a ranked list, but does not make a decision. Its purpose is to structure reasoning.

Consultants remain responsible for interpreting results, validating contextual assumptions, and assembling final advisory recommendations. As mentioned before, The highest-ranked intervention only represents the strongest overall balance between relevance and feasibility found in the database. It does not represent the only viable option. Lower-ranked interventions may still be relevant depending on project-specific constraints such as budget or implementation speed; this is up to the consultants to interpret and decide on.

6 Why the Scores Are Trustworthy

Trust in advisory tools such as this one depends primarily on clarity of reasoning, rather than algorithmic complexity. The credibility of the recommender derives from the consistent evaluation of all interventions under the same framework and from the visibility of underlying assumptions.

Each criterion corresponds directly to a practical implementation consideration familiar to stakeholders. Because weights and scores remain accessible, users can understand how outcomes change when priorities shift. This transparency enables constructive discussion instead of blind reliance on model outputs.

The scoring model follows established principles from multi-criteria decision-making literature, where alternatives are evaluated across clearly defined dimensions and aggregated using transparent weighting schemes (Ciardiello and Genovese 2023; Ezell, Lynch, and Hester 2021).

Each dimension reflects a real-world aspect of implementation. Impact reflects expected outcomes, effort reflects complexity, risk reflects uncertainty, cost reflects financial burden, and feasibility reflects practical executability.

Linear aggregation models are known to produce stable rankings when weight variations remain moderate (Ezell, Lynch, and Hester 2021; Cakira, Hecht, and Krellenberg 2021). This robustness ensures that recommendations are not excessively sensitive to minor preference changes. The model therefore supports reliable decision structuring even under uncertainty.

An illustrative example demonstrates this principle. A neighborhood activity program may offer limited measurable impact but require minimal investment and organizational effort. The recommender reflects this trade-off explicitly, allowing stakeholders to recognize its value as a feasible short-term improvement rather than dismissing it due to modest impact alone.

The model's strength lies in exposing reasoning rather than asserting certainty.

7 Recommendations

The recommender should be positioned as an integral step in the consultancy workflow immediately following completion of the well-being impact scan. At this stage, analytical insight is fresh, and the

need for structured decision-making is greatest. Embedding the tool here ensures that recommendations consistently emerge from the same evaluative logic.

Successful adoption requires organizational embedding rather than purely technical deployment. Consultants must view the recommender as an extension of their expertise rather than a constraint on professional judgment. Training should therefore focus on interpretation of rankings and discussion of trade-offs rather than mechanical use of the tool. However, use of the tool should not be fully neglected. Proper adjustment of the model's weights allows it to properly reflect project-specific priorities and considerations.

Long-term value depends on active maintenance of the intervention database. Scores should be reviewed periodically, informed by lessons learned during project implementation. Over time, this transforms the system into a repository of organizational knowledge, allowing Heijmans to accumulate experience across projects instead of rediscovering insights repeatedly.

The transparency of the approach also provides strategic benefits in stakeholder communication. Municipalities and project partners increasingly expect evidence-based justification for development decisions. The recommender enables consultants to demonstrate how priorities such as affordability, feasibility, or long-term impact influence recommendations, strengthening credibility in collaborative planning processes (Afshari, Vatanparast, and Ćoćkalo 2016; Romero-Torres, Casado-Sanz, and Guirao 2022; Albán-Pérez, Serrano-Guzmán, and Pérez-Ruiz 2023).

8 Limitations

The system operates under several important limitations. Because intervention scores are derived from expert judgment rather than empirical outcome data, the model does not predict real-world effectiveness. Instead, it compares alternatives based on informed expectations. If the database is properly maintained, with the scores updated by experts at Heijmans following knowledge gained by completed projects (past and present), the scores will gradually become more refined over time, better aligning with real-world findings. In its current as-delivered state, the scores are grounded but still speculative.

The model assumes independence between evaluation criteria, meaning interaction effects between interventions or contextual factors are not explicitly modelled. In practice, urban interventions often influence one another, and such dynamics remain outside the scope of the current system.

Results are also sensitive to chosen weights. While flexibility is necessary to reflect project priorities, inappropriate weight selection may lead to misleading rankings if not guided by professional expertise. For this reason, consultant interpretation remains essential.

These limitations reinforce the intended role of the recommender: it augments professional judgment but does not replace it. Future development could incorporate empirical validation through post-project evaluation, gradually strengthening the evidence base underlying intervention scores.

9 Conclusion

This project addresses a key consultancy challenge: translating analytical insight into actionable recommendations. While Heijmans already possesses strong diagnostic capabilities through the well-being impact scan, the transition from analysis to intervention selection lacked structure and transparency.

The proposed recommender introduces a systematic yet flexible framework that externalizes decision reasoning. By making trade-offs explicit, it improves consistency across consultants, accelerates advisory workflows, and enhances explainability toward stakeholders.

Additionally, the system enhances organizational learning. Each use contributes to a growing knowledge base that can refine future decisions, allowing Heijmans to scale data-driven consultancy without sacrificing contextual sensitivity.

The recommender therefore represents not merely a technical tool but a methodological advancement in

how well-being insights are translated into practical urban interventions.

It provides a foundation for scaling data-driven consultancy while maintaining flexibility for context-specific decision-making, forming a valuable addition to (but not replacement of) the well-being impact scan process.

Appendix A: Overlap Analysis — Excluded Interventions

Buurtvoorzieningenhub was excluded on the basis of strong conceptual overlap with INT_042 Buurthub met dagelijkse basisvoorzieningen. Both interventions describe a neighbourhood-level hub providing daily basic services to residents. The mechanism, target group, and intended outcome are equivalent. The Heijmans entry does not introduce a meaningfully distinct intervention type.

Buurtbudget was excluded on the basis of strong conceptual overlap with INT_009 Bewonersbudget en micro-subsidies voor wijkinitiatieven. Both interventions describe a participatory budget mechanism through which residents receive financial resources to initiate neighbourhood activities. The underlying empowerment logic and delivery mechanism are identical.

Autoluwe straat was excluded as a sub-component of INT_025 Parkeren en autoluw ontwerp sturen. The concept of a car-light street is explicitly embedded within the broader intervention of car-light design and parking management. Adding it as a separate entry would create a redundant, narrower version of an intervention already present in the database.

Parkeren op afstand van de voordeur was excluded as a sub-component of INT_025 Parkeren en autoluw ontwerp sturen. Relocating parking away from front doors is one specific implementation measure within the broader car-light design intervention and does not constitute a distinct intervention in its own right.

Buurthuis was excluded as a sub-component of INT_043 Inclusieve multifunctionele wijkvoorziening. The buurthuis represents a more generic and less developed version of the inclusive multifunctional neighbourhood facility already present in the database. INT_043 encompasses the buurthuis concept while adding universal access design, inclusive programming, and mixed community use as defining characteristics.

Schaduwrijke plekken was excluded as a sub-component of INT_018 Hittestress aanpakken. The creation of shaded outdoor spaces is one physical measure within the broader heat stress intervention. It does not represent a standalone intervention with a distinct mechanism or outcome logic.

Fietsroutes was excluded on the basis of substantial overlap with INT_024 Fiets- en looproutes met deelmobiliteitspunten. The Heijmans entry describes cycling infrastructure in general terms, which is already captured within the existing intervention. Retaining both would introduce redundancy within the same thematic and mechanistic space.

Toegankelijke en veilige (door)fietsroutes was excluded for the same reasoning as Fietsroutes. Although the safety and accessibility framing adds a minor nuance, the intervention remains within the same mechanistic and thematic territory as INT_024 and does not constitute a sufficiently distinct entry to justify inclusion alongside it.

Buurt, pluk- of moestuin was initially flagged as a strong overlap with INT_003 Lokale voedsel- en moestuininitiatieven ondersteunen but was subsequently reinstated. Upon further review, INT_003 is a programmatic intervention to support and stimulate community garden initiatives, whereas Buurt, pluk- of moestuin refers to the physical infrastructure itself. These represent distinct intervention types operating at different levels and were therefore treated as non-overlapping.

Beheervereniging was initially flagged as overlapping with INT_026 Beheervereniging voor voedselbos of boomgaard but was subsequently reinstated. INT_026 is explicitly scoped to the management of a food forest or orchard, whereas Beheervereniging in the Heijmans context refers to a general residents management association with broader scope. These are sufficiently distinct to warrant separate entries.

Appendix B: Scoring Rationale - Initial Batch

INT_001 — Beweegprogramma voor bewoners

A neighbourhood physical activity programme has modest direct impact on wellbeing at area level; it helps but does not move the needle dramatically on its own, hence impact 2. It is however one of the easiest interventions to execute: low effort, low risk, and very cheap to run, typically delivered through existing welfare or GGD partners with minimal budget. Measurability is reasonably good through participation rates and GGD health monitors, and data to support it is available at national and regional level. Feasibility is scored at its highest; this is a well-proven, widely used intervention type in Dutch neighbourhoods with clear delivery partners and no political friction. Scores: impact 2, effort 1, risk 1, cost 1, measurability 4, data availability 3, feasibility 5.

INT_008 — Welzijn op recept

Welzijn op recept has a moderate impact score of 3; it addresses root causes of health complaints like loneliness and inactivity effectively at the individual level, but its neighbourhood-wide impact is limited by how many people actually get referred and follow through. Effort and risk are both low: the intervention runs through existing GP and welfare infrastructure, requires no physical construction, and the model is well tested in the Netherlands. Cost is scored 1 because it does not require a significant budget beyond the welzijnsmakelaar capacity that many municipalities already fund. Measurability and data availability are both 4; referral rates, follow-through and health outcomes can be tracked through GP and welfare registration systems, and GGD monitors provide solid baseline data. Feasibility lands at 4 rather than 5 because implementation quality varies significantly by region; smaller municipalities without an established welzijnsorganisatie network will struggle to run this effectively. Scores: impact 3, effort 2, risk 1, cost 1, measurability 4, data availability 4, feasibility 4.

INT_019 — Groenblauwe maatregelen voor hittestress, wateropvang en biodiversiteit

Groenblauwe maatregelen score high on impact because they address multiple pressing urban challenges simultaneously: heat stress, water retention and ecological quality. The effect is durable and area-wide, not just individual. Effort and cost are both scored 4 because these are physical infrastructure interventions requiring planning, municipal coordination, contractor involvement and multi-year implementation waves. Risk is moderate at 3, as the technical risks are manageable but coordination between municipality, developer and water authorities adds complexity. Measurability and data availability are 3; temperature and water retention can be measured but attribution to specific interventions is not always straightforward. Feasibility lands at 4 because while the measures are well understood, getting all required parties aligned and funded in a Dutch development context takes significant time and political will, but is certainly achievable. Scores: impact 4, effort 4, risk 3, cost 4, measurability 3, data availability 3, feasibility 3.

INT_025 — Parkeren en autoluw ontwerp sturen

The direct wellbeing impact of car-light design is scored 2; while reducing car dependency has long-term benefits for air quality, safety and liveability, these effects are diffuse and slow to materialise at neighbourhood level. The intervention scores high on effort at 4 because it requires redesigning public space, relocating parking infrastructure and coordinating between municipality, developer and residents throughout a lengthy planning process. Risk is also scored 4, and this is where the political dimension becomes critical: autoluwe wijken are increasingly contested in Dutch municipal politics. Right-leaning municipalities, which are common outside the four major cities, are often explicitly resistant to car-light policies, viewing them as imposing lifestyle choices on residents who depend on cars for work and daily life. This means the same intervention that proceeds smoothly in Amsterdam or Utrecht can face serious opposition or outright rejection elsewhere. Cost is moderate at 3; parking restructuring requires physical works but is not the most expensive intervention in the set. Feasibility is scored low at 2 precisely because of this political variability: without municipal buy-in the intervention simply cannot happen, and that buy-in cannot be assumed across Heijmans' project portfolio which spans a wide range of Dutch municipalities. Scores: impact 2, effort 4, risk 4, cost 3, measurability 3, data availability 3, feasibility 2.

INT_029 — Financiële weerbaarheid versterken

Financiële weerbaarheid scores high on impact at 4 because financial stress is one of the most documented root causes of broader wellbeing problems in Dutch neighbourhoods. Households struggling with debt or income insecurity show significantly worse outcomes on health, social cohesion and mental wellbeing, meaning this intervention has a multiplier effect across multiple themes. Effort and risk are both scored low at 2; the intervention runs through existing municipal schuldhulpverlening infrastructure and welzijnsorganisaties, requires no physical works, and the model is well established across Dutch municipalities. Cost is also 2 because delivery largely piggybacks on publicly funded systems that already exist. Measurability is scored 4 because financial stress indicators, debt registration and uptake of debt help are all trackable through municipal systems, though granularity at neighbourhood level can be patchy depending on the municipality. Feasibility is 4 rather than 5 because effective delivery requires active collaboration between the municipality, welzijnsorganisatie and sometimes woningcorporaties; that coalition is achievable but not automatic, and in some regions waiting lists for schuldhulpverlening are already long. Scores: impact 4, effort 2, risk 2, cost 2, measurability 4, data availability 3, feasibility 4.

INT_038 — Rookvrije schoolomgeving

A smoke-free school environment scores moderate on impact at 3 because its effect is real but geographically and demographically limited to the school zone and its direct users. Where it excels is on every operational dimension: effort, risk and cost are all 1 because it is essentially a policy decision backed by signage and communication, requiring no infrastructure or significant budget. Measurability and data availability are both 5 because compliance is directly observable and GGD school health monitoring already tracks smoking behaviour around schools. Feasibility is 5 because the Dutch government actively supports rookvrije schoolterreinen and most municipalities and school boards are already aligned on this. Scores: impact 3, effort 1, risk 1, cost 1, measurability 5, data availability 5, feasibility 5.

INT_041 — Zichtbaar wijkbeheer en aanspreekpunt

A visible neighbourhood manager with a clear contact point has high impact because it directly addresses perceived safety and management quality, which are consistently among the top concerns in Dutch neighbourhood surveys. Effort and cost are low since it essentially means deploying one person in a structured role with no physical works required. Risk is scored 1 because this is one of the most straightforward interventions to execute with minimal things that can go wrong. Feasibility is 5 because every municipality has the infrastructure to implement this and there are no political or technical barriers worth mentioning. Scores: impact 4, effort 2, risk 1, cost 2, measurability 3, data availability 3, feasibility 5.

Appendix C: Scoring Rationale - Heijmans Interventions

INT_045 — Bewoners werk/ontmoetingsruimte

A shared residents workspace and meeting space has moderate impact on social cohesion by providing a consistent physical anchor for community activity. Effort and cost are both scored 3 because the intervention requires a dedicated space, basic furnishing and ongoing management, which typically falls to a woningcorporatie or developer. Risk is low at 2 since the concept is well understood in Dutch neighbourhood development. Measurability and data availability are both scored 2 because usage and social outcomes are trackable in principle but rarely systematically measured in practice. Feasibility is 3 reflecting that while the concept is viable, securing and maintaining a dedicated space requires sustained commitment from the lead actor. Scores: impact 3, effort 3, risk 2, cost 3, measurability 2, data availability 2, feasibility 3.

INT_046 — Buurt, pluk- of moestuin

A community or food picking garden has modest neighbourhood-wide impact, primarily benefiting the direct participants rather than the broader resident population. Effort and cost are both low at 2 since these initiatives typically build on existing green space and are largely resident-driven with minimal professional input required. Risk is scored 1 as there are virtually no failure modes that create negative consequences for the neighbourhood. Measurability and data availability are 2 because while participation

can be tracked, connecting garden activity to measurable wellbeing outcomes is methodologically difficult. Feasibility is 4 because the format is widely supported by Dutch municipalities and requires no complex approvals or large budgets. Scores: impact 2, effort 2, risk 1, cost 2, measurability 2, data availability 2, feasibility 4.

INT_047 — Co-creatie

Co-creation processes involving residents in design or planning decisions have moderate impact when they lead to genuinely implemented outcomes, as they increase ownership and alignment between built environment and resident needs. Effort is low at 2 and cost is 1 since co-creation is primarily a process intervention requiring facilitation rather than physical resources. Risk is scored 3 because co-creation processes that raise resident expectations and then fail to translate into visible change can seriously damage trust between residents and the developer or municipality, a well-documented failure pattern in Dutch participatietrajecten. Measurability is 2 since the outcomes of co-creation are diffuse and long-term. Feasibility is 4 as the method is widely used and accepted in Dutch urban development practice. Scores: impact 3, effort 2, risk 3, cost 1, measurability 2, data availability 2, feasibility 4.

INT_048 — Geclusterd wonen ouderen/Knarrenhof

Clustered senior housing models such as the Knarrenhof concept have high impact on healthy ageing, loneliness reduction and social cohesion among older residents, addressing one of the most pressing demographic challenges in Dutch neighbourhoods. Effort is scored 4 and cost 5 because this is a full housing development intervention requiring architectural design, construction, and long-term management agreements between woningcorporatie and developer. Risk is moderate at 3 due to the complexity of aligning housing typology, care infrastructure and resident selection processes. Measurability and data availability are both 3 as health and loneliness outcomes among older residents can be tracked through GGD monitors and resident surveys, though attribution to the housing model specifically requires careful study design. Feasibility is 3 reflecting that while the concept is proven in the Netherlands, land availability and developer commitment are not guaranteed in all project contexts. Scores: impact 4, effort 4, risk 3, cost 5, measurability 3, data availability 3, feasibility 3.

INT_049 — Gemeenschappelijke (binnen)tuin

A shared courtyard or inner garden has moderate impact on social cohesion by creating informal meeting opportunities between residents at building or block level. Effort and cost are both 3 since this requires physical design and construction as part of the development, though it is a standard element in Dutch residential projects. Risk is low at 2 as it is a well-understood physical intervention with predictable outcomes. Measurability and data availability are both 2 because measuring the social impact of shared green space at building level is not systematically done in Dutch practice. Feasibility is 4 since shared gardens are a common feature of woningcorporatie and developer projects and face no significant barriers to implementation. Scores: impact 3, effort 3, risk 2, cost 3, measurability 2, data availability 2, feasibility 4.

INT_050 — Kunst in de leefomgeving

Public art in the living environment has minimal measurable impact on core wellbeing dimensions such as health, safety or social cohesion, functioning primarily as an aesthetic and identity intervention. Effort and cost are both low at 2 since commissioning and placing artworks does not require significant coordination or infrastructure. Risk is 1 as there are almost no negative consequences associated with this intervention. Measurability and data availability are both scored 1 because there is no established method or data infrastructure to measure wellbeing impact of public art at neighbourhood level in the Netherlands. Feasibility is 4 as municipalities and developers regularly commission public art as part of area development with no structural barriers. Scores: impact 1, effort 2, risk 1, cost 2, measurability 1, data availability 1, feasibility 4.

INT_051 — Natuurlijke speelaanleidingen

Natural play elements in public space have moderate impact on children's physical activity, motor development and outdoor play behaviour, contributing meaningfully to the gezonde leefstijl theme. Effort is

scored 2 and cost 2 since these elements are typically simple, low-maintenance installations integrated into green space design. Risk is 1 as this is one of the safest and most straightforward physical interventions in the set. Measurability and data availability are both 3 since children's physical activity and outdoor play can be tracked through observation studies and GGD youth health monitors. Feasibility is 5 because natural play elements are universally supported by municipalities, schools and developers and face no political or technical barriers. Scores: impact 3, effort 2, risk 1, cost 2, measurability 3, data availability 3, feasibility 5.

INT_052 — Voortuin, veranda, (Delftse) stoep, zitnis

Front gardens, porches, Delft-style stoeps and sitting niches have well-documented positive effects on informal social contact at street level, contributing to social cohesion through passive interaction between neighbours. Effort is scored 1 as these are standard architectural design elements requiring no additional coordination beyond the design process. Risk is 1 and cost 2 since these are low-cost design choices with no meaningful failure modes. Measurability and data availability are both 2 because while street-level social interaction is observable, systematic measurement at neighbourhood level is not standard practice. Feasibility is 5 as these elements are embedded in Dutch residential design tradition and face no barriers to implementation. Scores: impact 3, effort 1, risk 1, cost 2, measurability 2, data availability 2, feasibility 5.

INT_053 — Talent/werkervaring programma

Talent and work experience programmes for young people have moderate impact on individual development, social mobility and connection to the labour market, addressing the opgroeien en ontwikkelen theme directly. Effort is 3 and cost 3 because these programmes require sustained coordination between gemeente, onderwijs and private employers, which takes meaningful time and resources to establish and maintain. Risk is 2 as the intervention is well-established in Dutch youth work practice with limited downside risk. Measurability and data availability are both 3 since participation rates, employment outcomes and educational trajectories can be tracked through gemeentelijke registratie and ROC data. Feasibility is 3 reflecting that while the concept is proven, securing employer partners and maintaining programme quality over time requires consistent effort. Scores: impact 3, effort 3, risk 2, cost 3, measurability 3, data availability 3, feasibility 3.

INT_054 — Hittebestendige woning

Heat-resistant housing design has moderate to significant impact on resident health and comfort as Dutch summers become increasingly extreme, directly addressing the milieukwaliteit en klimaatadaptatie theme. Effort is scored 4 and cost 4 because heat-resistant design requires specific building specifications, materials and potentially mechanical systems, all of which add to construction cost and complexity. Risk is 3 reflecting that while the technical solutions are available, integrating them into standard development workflows requires deliberate attention to avoid being value-engineered out. Measurability and data availability are both 4 as indoor temperature, energy consumption and comfort can be directly measured and KNMI climate data provides strong baseline. Feasibility is 3 because while increasingly required by building regulations, full heat-resistant design standards are not yet uniformly applied across Dutch development projects. Scores: impact 3, effort 4, risk 3, cost 4, measurability 4, data availability 4, feasibility 3.

INT_055 — Beheervereniging

A management association for shared spaces has moderate impact on social cohesion by creating a formal structure for residents to collectively take responsibility for their shared environment. Effort is 2 and cost 1 since establishing a beheervereniging is primarily an organisational and legal process requiring minimal financial resources. Risk is 2 as the main failure mode is low resident participation, which is manageable through good onboarding. Measurability and data availability are both 2 since association activity and membership can be tracked but wellbeing impact is indirect and difficult to attribute. Feasibility is 4 as this is a well-established legal structure in Dutch housing law with clear formation procedures. Scores: impact 3, effort 2, risk 2, cost 1, measurability 2, data availability 2, feasibility 4.

INT_056 — (Bio)materialen voor hogere akoestische kwaliteit

Using bio-based or high-performance acoustic materials in construction has moderate impact on resident wellbeing by reducing noise nuisance, which is consistently linked to stress and sleep disturbance. Effort is 4 and cost 4 because sourcing, specifying and applying acoustic bio-materials requires specialist knowledge and adds meaningful cost to the construction budget. Risk is 3 as performance depends heavily on correct installation and the materials market for bio-acoustic solutions is still maturing. Measurability and data availability are both 4 since acoustic performance can be precisely measured through standardised tests and noise complaint data is tracked by municipalities. Feasibility is 3 as the materials are available but not yet standard practice in Dutch construction, requiring deliberate specification by the developer. Scores: impact 3, effort 4, risk 3, cost 4, measurability 4, data availability 4, feasibility 3.

INT_057 — Bikescout

A cycling safety scout programme for primary school children has modest neighbourhood-wide impact but delivers meaningful benefits for the specific target group by improving cycling confidence and traffic safety awareness. Effort is 2 and cost 1 since the programme runs through schools with minimal resource requirements beyond trained instructors. Risk is 1 as this is a proven, low-complexity programme with no meaningful failure modes. Measurability and data availability are both 3 as participation and cycling proficiency can be tracked through school records and GGD youth health monitors include cycling behaviour indicators. Feasibility is 5 since bikescout-type programmes are actively promoted by Veilig Verkeer Nederland and widely implemented across Dutch primary schools. Scores: impact 2, effort 2, risk 1, cost 1, measurability 3, data availability 3, feasibility 5.

INT_058 — Samenwerking onderwijs

Education collaboration between schools and neighbourhood organisations has moderate impact on child development outcomes and the connection between school and living environment, directly addressing the opgroeien en ontwikkelen theme. Effort, cost, risk and feasibility are all scored at 3 reflecting that this is a genuinely mid-range intervention; it is beneficial when it works but requires sustained coordination between multiple parties that do not always share the same priorities or timelines. Measurability and data availability are both 3 as educational outcomes, school-neighbourhood programme participation and youth wellbeing indicators are all tracked through bestaande registratiesystemen but attribution to the collaboration specifically is not straightforward. Scores: impact 3, effort 3, risk 3, cost 3, measurability 3, data availability 3, feasibility 3.

INT_059 — Sportdagen, buurtfeesten, koffiegroepjes

Sports days, neighbourhood events and coffee groups have modest impact at neighbourhood level, primarily creating incidental social contact rather than sustained social cohesion. Effort is 2 and cost 2 since these events are typically organised through welzijnsorganisaties and bewonersinitiatieven with low resource requirements. Risk is 1 as there are no meaningful negative consequences associated with this intervention type. Measurability and data availability are both 2 since attendance can be tracked but sustained social impact is not measurable through standard Dutch data sources. Feasibility is 5 as these activities face no barriers whatsoever and can be organised in virtually any neighbourhood context. Scores: impact 2, effort 2, risk 1, cost 2, measurability 2, data availability 2, feasibility 5.

INT_060 — Buurthoreca

A neighbourhood catering facility has moderate impact on social cohesion and access to voorzieningen when it functions well, providing a daily social anchor for residents. Effort is 4 and cost 4 because establishing a commercial food and beverage operation requires significant upfront investment, permits, fit-out and ongoing operational management. Risk is scored 5 because buurthoreca concepts face a very high failure rate in Dutch neighbourhood contexts; commercial viability is genuinely uncertain, and many concepts close within two years due to insufficient footfall, high fixed costs, and competition from mainstream hospitality. Measurability and data availability are both 2 since commercial viability can be tracked but social impact measurement is harder. Feasibility is scored 1 because without a proven operator, a viable business model and sustained municipal or corporatie support, this intervention will not succeed and should only be recommended where those conditions are demonstrably in place. Scores: impact 3, effort 4, risk 5, cost 4, measurability 2, data availability 2, feasibility 1.

INT_061 — Ecologische verbindingszones

Ecological connection zones linking green areas have moderate impact on environmental quality, biodiversity and resident wellbeing through improved access to nature. Effort is 4 and cost 4 because these zones require land planning, ecological design, coordination with waterschap and often multi-year phased implementation. Risk is 3 as coordination between gemeente, waterschap and developer adds complexity and potential for misalignment. Measurability and data availability are both 4 since biodiversity indicators, green coverage and ecological connectivity can be measured through established monitoring frameworks. Feasibility is 3 reflecting that while the concept is supported in Dutch spatial planning policy, the required multi-party coordination is not always achievable within standard development timelines. Scores: impact 3, effort 4, risk 3, cost 4, measurability 4, data availability 4, feasibility 3.

INT_062 — Groene buurtactivering

Green neighbourhood activation, mobilising residents to actively engage with and maintain green spaces, has moderate impact on both environmental quality and social cohesion through shared stewardship. Effort is 2 and cost 2 since this primarily requires community organisation and facilitation rather than physical infrastructure. Risk is 2 as the main risk is low resident uptake, which is manageable. Measurability and data availability are both 3 since green space quality and resident participation in green management can be tracked through gemeentelijke groenmonitoring. Feasibility is 4 as the intervention is well-supported by Dutch municipalities and requires no complex approvals. Scores: impact 3, effort 2, risk 2, cost 2, measurability 3, data availability 3, feasibility 4.

INT_063 — Meergeneratie wonen

Multi-generational living developments have high impact because they simultaneously address loneliness among elderly residents, housing affordability for younger residents, and social cohesion across demographic groups. Effort is 4 and cost 4 because designing and managing a mixed-age residential development requires deliberate architectural choices and active community management beyond standard housing programmes. Risk is 3 as managing the social dynamics of mixed-age communities requires ongoing facilitation and clear governance structures. Measurability and data availability are both 3 since social cohesion and health outcomes across age groups can be tracked through GGD monitors and resident surveys. Feasibility is 3 reflecting that while the concept is increasingly supported in Dutch housing policy, it requires deliberate commitment from developers and woningcorporaties who are not always willing to take on the additional management complexity. Scores: impact 5, effort 4, risk 3, cost 4, measurability 3, data availability 3, feasibility 3.

INT_064 — Mobiliteitshub

A mobility hub combining shared transport options, bicycle parking and public transport access has moderate impact on mobility and access to voorzieningen, particularly for residents without private cars. Effort is 4 and cost 4 because mobility hubs require physical infrastructure, coordination with vervoerders and integration into the wider transport network. Risk is 3 as hub viability depends on sufficient user adoption and transport partner commitment, both of which carry uncertainty. Measurability and data availability are both 4 since mobility usage, public transport ridership and shared vehicle utilisation can be tracked through established transport monitoring systems. Feasibility is 3 as hub implementation requires active involvement of gemeente and vervoerders whose priorities do not always align with developer timelines. Scores: impact 3, effort 4, risk 3, cost 4, measurability 4, data availability 4, feasibility 3.

INT_065 — Park

A neighbourhood park has high impact across multiple wellbeing themes including physical activity, mental health, social cohesion and environmental quality, making it one of the most broadly beneficial interventions in the entire set. Effort is scored 5 because creating a park from scratch requires land acquisition or allocation, full landscape design, construction, ecological planting and long-term maintenance infrastructure. Cost is 4 reflecting the substantial investment required for quality park design and construction. Risk is low at 2 as parks are universally valued and face minimal social or political opposition once funding and land are secured. Measurability and data availability are both 3 since green space access, usage frequency and health outcomes can be measured but attribution to a specific park requires

dedicated study. Feasibility is 3 because while parks are broadly supported, securing land and sustained municipal maintenance budget is not guaranteed in all development contexts. Scores: impact 5, effort 5, risk 2, cost 4, measurability 3, data availability 3, feasibility 3.

INT_066 — Label Gezonde Woning

A healthy home label certifying that a dwelling meets defined standards for air quality, acoustics, light and thermal comfort has moderate impact by incentivising healthier building standards and raising resident awareness. Effort is 3 and cost 2 since the label is primarily a policy and certification process rather than a physical intervention, though it requires design adjustments to meet criteria. Risk is 3 as certification processes can be gamed or lose credibility if not rigorously enforced. Measurability and data availability are both 4 since the label criteria are based on measurable building performance indicators and building performance data is increasingly available through national registratiesystemen. Feasibility is 3 as adoption depends on developer willingness and no mandatory national standard currently exists, though the concept is gaining traction in Dutch building practice. Scores: impact 3, effort 3, risk 3, cost 2, measurability 4, data availability 4, feasibility 3.

INT_067 — Cooperatieve vereniging

A cooperative association giving residents collective ownership and governance over shared assets has moderate impact when functioning well, building strong social cohesion and long-term community resilience. Cost is 1 as cooperatives are primarily organisational structures requiring minimal financial investment to establish. Risk is scored 5 because cooperative associations in Dutch neighbourhoods have a well-documented history of governance conflicts, free-rider problems and organisational collapse, particularly in diverse communities with heterogeneous interests and limited governance experience. Measurability is 1 because measuring whether a cooperative structure actually improves wellbeing is extremely difficult. Feasibility is 3 as the legal structure exists and is supported in Dutch housing law, but success depends heavily on strong community leadership which cannot be assumed. Scores: impact 3, effort 3, risk 5, cost 1, measurability 1, data availability 2, feasibility 3.

INT_068 — Tijdelijke huur voor sociale of culturele voorzieningen

Temporarily renting spaces at below-market rates for social or cultural uses has moderate impact by enabling community organisations and cultural initiatives to establish themselves during the development phase of a neighbourhood. Effort is 2 and cost 2 since this requires a management decision and lease agreement rather than physical construction. Risk is 3 because temporary arrangements create uncertainty for the organisations involved and there is a real risk that spaces are reclaimed for commercial use before social initiatives are self-sustaining. Measurability and data availability are both 2 as the impact of temporary social spaces is qualitative and not systematically tracked. Feasibility is 4 as this is a common practice in Dutch area development and faces no structural barriers beyond commercial pressure on the developer. Scores: impact 3, effort 2, risk 3, cost 2, measurability 2, data availability 2, feasibility 4.

INT_069 — Biodynamische verlichting

Biodynamic lighting that adjusts colour temperature and intensity to support the human circadian rhythm has moderate impact on resident health, sleep quality and mental wellbeing, particularly in high-density residential buildings. Effort is 3 and cost 3 because biodynamic lighting systems require specialist specification and are more expensive than standard lighting but are increasingly available through mainstream suppliers. Risk is 2 as the technology is mature and well-tested. Measurability and data availability are both 3 since sleep quality and circadian health outcomes can be measured through resident surveys and wearable devices, though systematic measurement in Dutch housing projects is not yet standard. Feasibility is 3 as the technology is available but requires deliberate inclusion in the building specification, which adds cost pressure in competitive development budgets. Scores: impact 3, effort 3, risk 2, cost 3, measurability 3, data availability 3, feasibility 3.

INT_070 — Buurtapp

A neighbourhood app facilitating communication and information sharing between residents has modest impact on social cohesion, primarily benefiting residents who are already inclined to engage with their

neighbourhood. Effort is 2 and cost 2 since app development and maintenance costs have fallen significantly and off-the-shelf platforms are widely available. Risk is 2 as apps rarely cause harm but frequently fail to achieve adoption beyond a small core of active users. Measurability and data availability are both 2 because while app usage statistics are trackable, connecting digital engagement to actual social cohesion outcomes is methodologically challenging. Feasibility is 4 as buurtapps are widely implemented across Dutch municipalities with no structural barriers. Scores: impact 2, effort 2, risk 2, cost 2, measurability 2, data availability 2, feasibility 4.

INT_071 — Buurthuiskamer

A neighbourhood living room providing an informal, low-threshold meeting space for residents has moderate impact on social cohesion and loneliness reduction, particularly for elderly and isolated residents. Effort is 2 and cost 2 since this requires a dedicated space and basic furnishing, typically provided by a woningcorporatie or welzijnsorganisatie. Risk is 1 as this is one of the most straightforward social infrastructure interventions with minimal failure modes. Measurability and data availability are both 2 because visitor numbers can be tracked but social impact measurement is not standard. Feasibility is 5 as buurthuiskamers are widely supported in Dutch welzijnsbeleid and can be implemented in virtually any neighbourhood context with willing partners. Scores: impact 3, effort 2, risk 1, cost 2, measurability 2, data availability 2, feasibility 5.

INT_072 — Buurtwinkels (maatschappelijke focus)

Neighbourhood shops with a social mission, combining retail with social services, employment for vulnerable residents or community functions, have moderate impact on access to voorzieningen and social inclusion. Effort is 3 and cost 3 because establishing a viable social enterprise retail concept requires business development support, a suitable space and ongoing operational management. Risk is 3 because socially-focused retail concepts face commercial viability pressures and frequently struggle to sustain themselves without ongoing subsidy. Measurability and data availability are both 2 since social impact of community retail is qualitative and not systematically measured in Dutch practice. Feasibility is 3 because success depends on finding a motivated operator and securing affordable retail space, which varies significantly by location. Scores: impact 3, effort 3, risk 3, cost 3, measurability 2, data availability 2, feasibility 3.

INT_073 — Community manager / Buurtcoach

A community manager or neighbourhood coach has high impact because a visible, trusted person connecting residents to each other and to services is one of the most consistently effective social cohesion interventions documented in Dutch neighbourhood practice. Effort is 2 and cost 2 since this is a personnel deployment rather than a physical or programmatic intervention. Risk is scored 1 because this intervention has virtually no chance of failure; even a mediocre community manager creates more connection than no community manager. Measurability and data availability are both 3 since social network formation, service referrals and resident satisfaction can be tracked through structured monitoring. Feasibility is 5 because every municipality and woningcorporatie has the organisational infrastructure to appoint a community manager, with no technical, political or financial barriers at the margin. Scores: impact 4, effort 2, risk 1, cost 2, measurability 3, data availability 3, feasibility 5.

INT_074 — Pumptrack

A pumptrack cycling circuit for children and youth has modest neighbourhood-wide impact but delivers real benefits for its target group through physical activity, skill development and a dedicated youth meeting place. Effort is 3 and cost 3 since pumptracks require dedicated land, civil engineering and a safety-compliant surface. Risk is 2 as the concept is well-proven in Dutch public space design. Measurability and data availability are both 3 since youth physical activity and facility usage can be tracked through observation and GGD youth monitors. Feasibility is 4 as pumptracks are increasingly popular in Dutch gemeente sportbeleid and face no significant barriers beyond land availability. Scores: impact 2, effort 3, risk 2, cost 3, measurability 3, data availability 3, feasibility 4.

INT_075 — Entreelobby

A well-designed entrance lobby contributes modestly to resident wellbeing by improving the first experience of returning home and creating a small informal meeting point. Effort is 3 and cost 3 since lobby design requires architectural attention and quality materials but is a standard element of residential construction. Risk is 2 as this is a contained physical intervention with predictable outcomes. Measurability is scored 1 because there is no established method or data to measure how lobby quality affects resident wellbeing, and data availability is also 1 for the same reason. Feasibility is 4 as this is entirely within the developer's control and requires no external approvals or partnerships. Scores: impact 2, effort 3, risk 2, cost 3, measurability 1, data availability 1, feasibility 4.

INT_076 — Flex homes (kangoeroe)

Flex homes or kangaroo housing, dwellings designed with an integrated secondary unit allowing elderly parents or other family members to live closely but independently, have high impact on intergenerational care, loneliness and housing flexibility. Effort is 5 and cost 5 because these dwellings require significant architectural adaptation, structural modifications and often planning adjustments that add substantially to development cost. Risk is 3 as market acceptance and planning permission for this non-standard typology carry real uncertainty. Measurability and data availability are both 3 since health, care burden and housing satisfaction outcomes can be measured. Feasibility is 3 reflecting that while the concept is supported in Dutch housing policy, few developers have integrated it into standard product lines. Scores: impact 5, effort 5, risk 3, cost 5, measurability 3, data availability 3, feasibility 3.

INT_077 — Hardlooproute

A dedicated or marked running route has modest neighbourhood-wide impact but meaningfully supports physical activity for residents who already run or are motivated to start. Effort is 2 and cost 2 since a running route requires only surface markings, wayfinding and potentially minor path improvements. Risk is 1 as this is one of the simplest physical interventions with no meaningful failure modes. Measurability and data availability are both 3 since running route usage can be tracked through Strava data and physical activity indicators are available through GGD monitors. Feasibility is 5 as running routes can be implemented in virtually any neighbourhood context with minimal budget and no planning barriers. Scores: impact 2, effort 2, risk 1, cost 2, measurability 3, data availability 3, feasibility 5.

INT_078 — Geclusterd en betaalbaar wonen (jongeren)

Clustered affordable housing specifically for young people addresses one of the most acute wellbeing and social equity challenges in the Netherlands, the near-complete exclusion of young adults from the housing market in most Dutch urban areas. Impact is scored 5 because secure, affordable housing for young people is a foundational condition for all other wellbeing dimensions. Effort is 5 and cost 5 because this is a full housing development intervention requiring land, construction financing, municipal housing allocation agreements and long-term management. Risk is 3 as the concept is proven but political prioritisation of youth housing varies between municipalities. Feasibility is scored 2 because delivering genuinely affordable clustered youth housing in the Netherlands currently depends on specific combinations of municipal land policy, woningcorporatie commitment and subsidies that are not reliably available in all project contexts. Scores: impact 5, effort 5, risk 3, cost 5, measurability 3, data availability 3, feasibility 2.

INT_079 — Leefgalerij

A leefgalerij, a wide shared gallery or outdoor corridor designed as a living space rather than just a circulation route, has moderate impact on social cohesion by creating informal meeting opportunities directly outside residents' front doors. Effort is 3 and cost 3 since this requires deliberate architectural design decisions that add width and therefore cost to the building envelope. Risk is 2 as the concept is well-understood in Dutch housing architecture. Measurability is scored 1 because measuring the actual social impact of a leefgalerij versus a standard corridor is extremely difficult and not done in standard Dutch housing evaluation practice. Feasibility is 4 as this is an architectural design choice within the developer's control that faces no external barriers. Scores: impact 3, effort 3, risk 2, cost 3, measurability 1, data availability 2, feasibility 4.

INT_080 — Luwte oase

A sheltered oasis, a protected outdoor space offering shade, wind shelter and a calm environment, has moderate impact on heat stress resilience and mental wellbeing by providing residents with accessible outdoor space that is comfortable in challenging weather conditions. Effort is 3 and cost 3 since this requires designed landscape elements including planting, shade structures and seating. Risk is 2 as the concept is straightforward and well-executed examples exist across Dutch cities. Measurability and data availability are both 3 since temperature and usage can be measured and KNMI climate data provides context. Feasibility is 4 as this is a standard element of quality public space design that faces no significant barriers. Scores: impact 3, effort 3, risk 2, cost 3, measurability 3, data availability 3, feasibility 4.

INT_081 — Neveltuin

A mist garden using water spray installations to cool public space has minimal measurable wellbeing impact beyond a short-term sensory experience, functioning primarily as a decorative and atmospheric element. Effort is 3 and cost 3 since mist installations require water infrastructure, maintenance and vandalism-resistant hardware. Risk is 2 as the concept is technically straightforward. Measurability is scored 1 because there is no established method to measure wellbeing impact of mist installations and the effect is too transient to meaningfully assess. Feasibility is 4 as the technology is available and increasingly used in Dutch public space design, though the limited impact makes it a low priority relative to other interventions. Scores: impact 1, effort 3, risk 2, cost 3, measurability 1, data availability 3, feasibility 4.

INT_082 — Kwiekroute

A kwiekroute, a designated walking route designed specifically for elderly residents with attention to safety, bench placement, surface quality and points of interest, has moderate impact on physical activity, loneliness and mental wellbeing among older residents. Effort is 2 and cost 2 since the route primarily requires thoughtful design and minor surface improvements rather than major construction. Risk is 1 as this is a very low-complexity intervention with no meaningful failure modes. Measurability and data availability are both 3 since elderly physical activity and walking behaviour can be tracked through GGD monitors and dedicated evaluation studies. Feasibility is 5 because kwiekroutes are actively supported by Nederlandse gemeenten as part of ouderenbeleid and face no structural barriers. Scores: impact 3, effort 2, risk 1, cost 2, measurability 3, data availability 3, feasibility 5.

INT_083 — Parkeerhub

A parking hub consolidating car parking at the edge of a neighbourhood rather than distributing it through streets has modest direct wellbeing impact but enables broader improvements to public space quality and pedestrian safety. Effort is 4 and cost 4 because parking hubs require substantial civil engineering, often including multi-storey structures, and complex traffic management planning. Risk is 3 as resident acceptance of consolidated parking and changes to car access patterns can be contentious. Measurability and data availability are both 3 since parking usage, traffic flows and public space quality can be measured through standard urban monitoring. Feasibility is 3 because parkeerhubs require strong municipal commitment and resident acceptance, which is not guaranteed particularly in municipalities with a car-oriented political culture. Scores: impact 2, effort 4, risk 3, cost 4, measurability 3, data availability 3, feasibility 3.

INT_084 — Sportplekken (jongeren)

Dedicated sports spots for youth, such as outdoor facilities designed for basketball, skateboarding, calisthenics or other youth-preferred activities, have moderate impact on physical activity and social gathering among the target group. Effort is 3 and cost 3 since these require dedicated land, quality surfaces and appropriate equipment. Risk is 2 as the concept is well-proven in Dutch public space design. Measurability and data availability are both 3 since youth facility usage and physical activity can be tracked through observation and GGD youth monitors. Feasibility is 4 as youth sports facilities are widely supported by Dutch municipalities and developers as a standard neighbourhood amenity. Scores: impact 3, effort 3, risk 2, cost 3, measurability 3, data availability 3, feasibility 4.

INT_085 — Veilige autovrije routes naar school/speeltuin

Safe car-free routes connecting homes to schools and playgrounds have high impact on child safety, active

mobility and parental wellbeing, addressing one of the most consistently raised concerns in Dutch neighbourhood resident surveys. Effort is 3 and cost 3 since routes require infrastructure adjustments including surface markings, traffic calming and signage. Risk is 2 as the concept is well-established and broadly supported. Measurability and data availability are both 4 since school route safety can be measured through traffic counts, accident data and school travel surveys, all of which are available through Dutch gemeentelijke data. Feasibility is 4 as safe school routes are actively supported by Veilig Verkeer Nederland and most municipalities, with dedicated subsidy programmes available. Scores: impact 4, effort 3, risk 2, cost 3, measurability 4, data availability 4, feasibility 4.

INT_086 — Thuis technologie

Home technology solutions supporting independent living for elderly and disabled residents, including smart sensors, fall detection and remote care monitoring, have moderate impact on health, safety and quality of life for the target group. Effort is 3 and cost 3 since technology deployment requires hardware installation, software integration and user training. Risk is 3 because technology adoption among elderly residents is often lower than anticipated and privacy concerns can create resistance. Measurability and data availability are both 3 since health outcomes, care burden and independent living duration can be tracked through zorgregistratie. Feasibility is 3 reflecting that while the technology is available, integrating it effectively into Dutch zorginfrastructuur requires coordination between developer, zorgorganisatie and gemeente that is not always straightforward. Scores: impact 3, effort 3, risk 3, cost 3, measurability 3, data availability 3, feasibility 3.

INT_087 — Verblijfsplekken (jongeren)

Dedicated hangout spots designed for youth, spaces that are explicitly welcoming to young people rather than spaces from which they are implicitly excluded, have moderate impact on youth wellbeing, sense of belonging and reduction of antisocial behaviour. Effort is 2 and cost 2 since these can be created through relatively simple design interventions in public space. Risk is 2 as the concept is well understood in Dutch youth and public space design practice. Measurability and data availability are both 2 since youth space usage can be observed but connecting it to measurable wellbeing outcomes is not standard. Feasibility is 5 because creating youth hangout spots faces no structural barriers and is supported by Dutch gemeenten as part of jongerenbeleid. Scores: impact 3, effort 2, risk 2, cost 2, measurability 2, data availability 2, feasibility 5.

INT_088 — Electrische bakfietsen

Shared electric cargo bikes have modest impact on sustainable mobility and access to transport options, particularly for families and residents without cars. Effort is 2 and cost 3 since this requires purchasing and maintaining a fleet and setting up a sharing system. Risk is 2 as the concept is technically straightforward and increasingly common in Dutch cities. Measurability and data availability are both 3 since usage can be tracked through the sharing platform and mobility data is available through national transport monitoring. Feasibility is 4 as electric cargo bike sharing is increasingly supported by Dutch municipalities and faces no structural barriers beyond procurement and operational setup. Scores: impact 2, effort 2, risk 2, cost 3, measurability 3, data availability 3, feasibility 4.

INT_089 — (Gevel)bankjes

Facade benches placed directly outside dwellings have minimal measurable neighbourhood-wide impact, functioning primarily as micro-scale design elements that marginally facilitate informal social contact at street level. Effort is 1 and cost 1 since facade benches are among the cheapest and simplest physical interventions possible, requiring only basic materials and placement. Risk is 1 as there are no meaningful failure modes. Measurability and data availability are both 2 because while social interaction at bench locations is observable, systematic impact measurement is not possible or warranted at this scale. Feasibility is 5 as facade benches can be placed anywhere with no approvals, permits or significant resources required. Scores: impact 1, effort 1, risk 1, cost 1, measurability 2, data availability 2, feasibility 5.

INT_090 — Bewonersbegeleiding

Resident guidance and support, structured accompaniment helping residents navigate services, resolve

conflicts and build self-sufficiency, has moderate impact on wellbeing by addressing practical barriers that prevent residents from accessing support they need. Effort is 2 and cost 2 since this is a professional support role that can be embedded within existing welzijnsorganisatie capacity. Risk is 2 as the main risk is insufficient caseload coverage or high turnover of support workers, both of which are manageable. Measurability and data availability are both 2 since referral outcomes and service uptake can be tracked but broader wellbeing impact is indirect and difficult to measure through standard Dutch data sources. Feasibility is 4 as bewonersbegeleiding is a well-established role in Dutch welzijnswerk with clear delivery infrastructure across most municipalities. Scores: impact 3, effort 2, risk 2, cost 2, measurability 2, data availability 2, feasibility 4.

Appendix D: Interventiecategorie to Database Category Mapping

The table below documents the mapping of each Heijmans Interventiecategorie value to the corresponding database category. This mapping was used to assign category identifiers to all 46 new interventions in the intervention-category-theme mapping table.

Table 2: Heijmans Interventiecategorie to Database Category Mapping

Heijmans Interventiecategorie	Database Category	Rationale
Verblijfsplekken buiten	CAT_05 Leefomgeving en klimaat	Outdoor spaces relate directly to living environment quality
Woongebouwen	CAT_06 Wonen en voorzieningen	Building-level interventions fall under housing and facilities
Gezamenlijke ruimten	CAT_02 Sociale cohesie	Shared spaces primarily serve social meeting and bonding functions
(Steden)bouwfysica	CAT_05 Leefomgeving en klimaat	Building physics interventions address acoustic, thermal and environmental quality
Digitale middelen	CAT_07 Governance en participatie	Digital tools primarily support communication and area governance
Buurtactivering	CAT_02 Sociale cohesie	Neighbourhood activation is directed at community connection and bonding
Buurtvoorzieningen	CAT_06 Wonen en voorzieningen	Neighbourhood facilities fall under accessible local provisions
Zeggenschap bewoners	CAT_07 Governance en participatie	Resident governance mechanisms are a core element of participation
Beweegroutes	CAT_04 Mobiliteit en toegankelijkheid	Movement routes concern active mobility and accessibility
Klimaatadaptatie	CAT_05 Leefomgeving en klimaat	Climate adaptation measures address environmental and living quality
Parkeren	CAT_04 Mobiliteit en toegankelijkheid	Parking infrastructure relates to mobility and area accessibility
Sportplekken	CAT_03 Gezondheid en welzijn	Sports facilities directly support physical health and active lifestyle

Heijmans Interventiecategorie	Database Category	Rationale
Co-creatie	CAT_07 Governance en participatie	Co-creation is a participatory process involving residents in governance
Kunst & Cultuur	CAT_08 Identiteit en cultuur	Arts and culture interventions address place identity and cultural programming
Speelplekken	CAT_03 Gezondheid en welzijn	Play spaces support physical development and health among children
Semi-publieke overgangszones	CAT_02 Sociale cohesie	Transitional zones between private and public space facilitate informal social contact
Talentontwikkeling	CAT_09 Opgroeien en ontwikkeling	Talent development is a core component of youth and developmental support
Talent-/werkervaringsprogramma	CAT_09 Opgroeien en ontwikkeling	Work experience programmes address youth development and labour market connection
Gevarieerd groen	CAT_05 Leefomgeving en klimaat	Varied green infrastructure relates to biodiversity and living environment quality
Placemaking	CAT_08 Identiteit en cultuur	Placemaking interventions address local identity and cultural character of an area
Overgangszones	CAT_02 Sociale cohesie	Transition zones between spaces facilitate informal social interaction
Deelmobiliteit	CAT_04 Mobiliteit en toegankelijkheid	Shared mobility directly concerns accessible and sustainable transport
Kleine ontmoetingsplekken	CAT_02 Sociale cohesie	Small meeting spots are designed to facilitate social contact and community bonding

References

Abdelli, A., A. Mokhtari, B. Beldjilali, and I. E. Tibermacine. 2020. "Dealing with Value Constraints in Decision Making Using MCDM Methods." *Journal of Computational Science* 44: 101153. https://doi.org/10.1016/j.jocs.2020.101153.

Afshari, A. R., M. Vatanparast, and D. Ćoćkalo. 2016. "Application of Multi Criteria Decision Making to Urban Planning: A Review." *Journal of Engineering Management and Competitiveness* 6 (1): 46–53.

Albán-Pérez, C. A., M. F. Serrano-Guzmán, and D. D. Pérez-Ruiz. 2023. "Weighted Sums Method Applied for Decision Making in Improvement Towards Complete Street: A Case Study in Popayan, Colombia." *Revista Legado de Arquitectura y Diseño* 18 (34).

Cakira, S., R. Hecht, and K. Krellenberg. 2021. "Sensitivity Analysis in Multi-Criteria Evaluation of the Suitability of Urban Green Spaces for Recreational Activities." *AGILE: GIScience Series* 2 (22). https://doi.org/10.5194/agile-giss-2-22-2021.

Ciardiello, F., and A. Genovese. 2023. "A Comparison Between TOPSIS and SAW Methods." *Annals of Operations Research* 325: 967–94. https://doi.org/10.1007/s10479-023-05339-w.

Data Consultancy Team. 2026. "Methodological White Paper: Constrained Weighted Sum Model for Urban Intervention Recommendations."

Domshlak, C., S. Prestwich, F. Rossi, K. B. Venable, and T. Walsh. 2006. "Hard and Soft Constraints for Reasoning about Qualitative Conditional Preferences." *Journal of Heuristics* 12 (4-5): 263–85. https://doi.org/10.1007/s10732-006-7071-x.

Ezell, B., C. J. Lynch, and P. T. Hester. 2021. "Methods for Weighting Decisions to Assist Modelers and Decision Analysts: A Review of Ratio Assignment and Approximate Techniques." *Applied Sciences* 11 (21): 10397. https://doi.org/10.3390/app112110397.

Khadra, A., M. Hugosson, J. Akander, and J. A. Myhren. 2020. "Development of a Weight Factor Method for Sustainability Decisions in Building Renovation: Case Study Using Renobuild." *Sustainability* 12 (17): 7194. https://doi.org/10.3390/su12177194.

Marinoni, O. 2006. "A Discussion on the Computational Limitations of Outranking Methods for Land-Use Suitability Assessment." *International Journal of Geographical Information Science* 20 (1): 69–87.

Romero-Torres, J., N. Casado-Sanz, and B. Guirao. 2022. "The Contribution of MCDM to SUMP: The Case of Spanish Cities During 2006–2021." *International Journal of Environmental Research and Public Health* 19 (1): 294.

Vagiona, D. G. 2021. "Comparative Multicriteria Analysis Methods for Ranking Sites for Solar Farm Deployment: A Case Study in Greece." *Energies* 14 (24): 8371. https://doi.org/10.3390/en14248371.